

Entanglement Swapping Sandbox

Implementation Brief

Artifact class: Theory, validation, and implementation companion to the scenario-screening sandbox.

Purpose: Document the physical model, Bell-state measurement construction, loss treatment, metric definitions, software structure, and test strategy supporting the simulator.

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Implementation Summary

This brief documents the theory and implementation rigor behind the **Entanglement Swapping Sandbox**. It records the explicit four-qubit model, the branch-resolved Bell-state projection logic, the parameterized misaligned Bell-state measurement basis, the simple link-loss treatment, the computed metrics, and the validation strategy used to keep the simulator compact, interpretable, and technically trustworthy.

Implementation principle. Every user-facing result in the sandbox is traceable to a physically meaningful quantity: source-state parameters, Bell-basis projection, loss factor, or entanglement metric. The implementation is compact, but not ad hoc.

Classical channel and the four-branch structure. Within the ideal symmetric Bell-state-measurement model, the four branches occur with equal probability, and the unconditioned remote state is maximally mixed before the classical measurement result is communicated. More generally, source mismatch, Bell-state-measurement misalignment, and loss produce unequal branch weights, and the unconditioned remote state is a mixed state rather than a known usable entangled resource. It is the classical message, bounded by c , that reveals which branch occurred and therefore which conditional entanglement the remote parties share. This is why the simulator reports branch-resolved results rather than a single averaged output: the branch index i is the classical message, and suppressing it would obscure the physical mechanism that prevents faster-than-light signaling.

Companion document: see the matching **Technical Brief** for artifact purpose, scenario-screening value, use cases, and positioning.

Scope and Modeling Philosophy

The sandbox is a minimal model rather than a full network stack. It is intended to capture the most important ingredients of a two-source / one-hub entanglement swapping scenario without introducing unnecessary framework complexity.

The implementation therefore emphasizes:

- analytic transparency,
- branch-resolved state updates,
- simple but explicit loss treatment,
- metric separation,
- and fast iteration in a notebook-friendly format.

The simulator operates at the level of pure states and projective measurements rather than density matrices and Kraus operators. This is sufficient for the current artifact objective: the sources are parameterized pure states, the BSM is a projective measurement, and loss enters as a scalar factor on valid-event probability rather than as an amplitude-damping channel on the quantum state. This keeps the implementation laptop-friendly—the core

objects are 4×4 complex matrices and standard linear algebra operations, and no special quantum libraries are required beyond NumPy. This makes the simulator suitable for explanation, scenario screening, and controlled extension.

Physical Model

The simulator uses four qubits labeled 1, 2, 3, 4. Qubits 1 and 2 form the first source pair; qubits 3 and 4 form the second source pair. The hub acts on qubits 2 and 3.

The source states are parameterized as

$$|\psi_A\rangle_{12} = \cos \alpha_A |00\rangle + e^{i\phi_A} \sin \alpha_A |11\rangle,$$

$$|\psi_B\rangle_{34} = \cos \alpha_B |00\rangle + e^{i\phi_B} \sin \alpha_B |11\rangle.$$

The global state is

$$|\Psi\rangle_{1234} = |\psi_A\rangle_{12} \otimes |\psi_B\rangle_{34}.$$

This parameterization is compact but expressive enough to study source mismatch, relative phase effects, and departure from the maximally entangled case.

Bell-State Measurement Model

The hub performs a projective measurement on qubits 2 and 3. In the ideal case, the Bell basis is

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

The code computes a branch probability p_m and corresponding conditional remote density matrix $\rho_{14|m}$ for each allowed branch m . This is implemented by constructing the projector on the measured subsystem and then performing a partial trace over qubits 2 and 3.

Theory-to-code mapping. Each displayed branch corresponds to an explicit projection on qubits 2, 3, followed by normalization and reduction to the remote subsystem 1, 4.

Misaligned BSM Construction

To study nonideal hub behavior, the ideal Bell basis is generalized into rotated even and odd sectors. The even sector is parameterized as

$$|B_{e1}\rangle = \cos \gamma_e |00\rangle + e^{i\varphi_e} \sin \gamma_e |11\rangle,$$

$$|B_{e2}\rangle = \sin \gamma_e |00\rangle - e^{i\varphi_e} \cos \gamma_e |11\rangle,$$

and the odd sector as

$$|B_{o1}\rangle = \cos \gamma_o |01\rangle + e^{i\varphi_o} \sin \gamma_o |10\rangle,$$

$$|B_{o2}\rangle = \sin \gamma_o |01\rangle - e^{i\varphi_o} \cos \gamma_o |10\rangle.$$

At the ideal point

$$\gamma_e = \gamma_o = 45^\circ, \quad \varphi_e = \varphi_o = 0,$$

the basis reduces to the standard Bell basis.

This construction provides a compact way to study Bell-state measurement misalignment without introducing a full optical interferometer model.

Loss Model

The current implementation uses a simple link-level loss factor applied to the valid branch probability:

$$p_{\text{valid}} = \eta_2 \eta_3 \xi_{\text{BSM}} p_{\text{cond}},$$

where η_2 and η_3 are transmission factors on the source-to-hub links and ξ_{BSM} is an effective Bell-state measurement efficiency.

This model is intentionally simple. It does not yet apply amplitude damping or detector nonidealities directly to the quantum state. Instead, it separates state quality from valid-event rate, which is appropriate for the current artifact objective.

Computed Metrics

For each branch, the simulator computes:

- conditional branch probability p_{cond} ,
- valid branch probability p_{valid} ,
- concurrence,
- negativity,
- Bell-state fidelities,
- best Bell-state match,
- entanglement yield,
- fidelity yield.

Under strong BSM misalignment or pronounced source imbalance, the conditional remote state becomes increasingly mixed and less Bell-like. In the extreme limit, concurrence and negativity can approach zero, providing a useful numerical test of entanglement degradation. These metrics are chosen to expose different aspects of performance:

- **Concurrence** captures pure two-qubit entanglement strength.
- **Negativity** provides a complementary entanglement diagnostic.
- **Bell fidelity** indicates closeness to a canonical Bell target.
- **Yield metrics** combine occurrence and usefulness.

Software Structure

The notebook is organized into three cells:

- **Cell 1:** tools, state construction, branch projection, metrics, helpers, and shared labels,
- **Cell 2:** user interface and snapshot computation,
- **Cell 3:** figures, heatmaps, and parameter summaries.

This separation was chosen for clarity. The numerical core is isolated from the user interface, while the plotting layer inherits parameters from the shared application state.

Validation and Testing Strategy

The sandbox is compact enough that validation can be explicit and interpretable. Recommended checks include:

- **Normalization check:** branch probabilities sum to unity in the ideal no-loss case.
- **Ideal Bell-point check:** at $\alpha_A = \alpha_B = 45^\circ$, ideal BSM parameters, and $\eta_2 = \eta_3 = \xi_{\text{BSM}} = 1$, all four Bell branches are recovered symmetrically.

- **Loss scaling check:** valid probabilities scale linearly with $\eta_2\eta_3\xi_{\text{BSM}}$.
- **Metric bounds check:** concurrence, negativity, and fidelities remain within physical bounds.
- **State-reduction check:** conditional remote density matrices are Hermitian and trace-normalized.
- **Strong-degradation check:** under severe BSM misalignment or source imbalance, concurrence and negativity should decrease toward zero, and Bell-state fidelities should reflect loss of a clear Bell-like target.

Implementation rigor. The most important tests are not only numerical but structural: probability conservation, ideal-limit recovery, metric bounds, and clear inheritance of UI parameters into reported figures.

Expected Limits and Sanity Cases

Several special points are especially useful for testing and interpretation:

Ideal symmetric point

When both sources are maximally entangled and the BSM is ideal, the simulator should return symmetric Bell-branch behavior.

Source imbalance

As α_A and α_B move away from one another, the branch weights and remote-state quality should become asymmetric.

BSM misalignment

As the BSM basis is rotated away from the ideal Bell point, Bell-fidelity and entanglement metrics should degrade in a controlled way.

Loss-only degradation

When only $\eta_2, \eta_3, \xi_{\text{BSM}}$ are reduced, conditional state quality should remain unchanged while valid-event probabilities decrease.

Known Limitations

The current model is intentionally minimal. It does not yet include:

- amplitude-damping channels on the traveling qubits,
- detector dark counts,
- multiphoton effects,
- arrival-time overlap or interference-visibility modeling,
- memory decoherence,
- chained or nested repeater operation.

These omissions are deliberate. The current sandbox is intended to stay compact and transparent. More detailed models can be layered on top later if needed.

Extension Paths

Natural next steps include:

- an interference-visibility or path-mismatch dial at the hub,
- more physical loss channels,
- branch filtering and acceptance policies,
- a linear repeater-chain extension,
- exportable screening summaries for technical specification support,
- explicit classical channel latency modeling and its contribution to the heralding timing budget.

These extensions can be added without changing the core logic of the current implementation.

Closing Statement

The implementation value of the Entanglement Swapping Sandbox lies in traceability: each output is tied to a clear physical assumption, a visible parameter, and a controlled computation. That makes the artifact suitable not only for communication, but also for technical scrutiny and extension.

The simulator is deliberately compact, but it is built to show theoretical and implementation rigor.

The physical context that motivates this rigor is concrete: the Qunnect/Cisco GothamQ demonstration (February 18, 2026) helped establish entanglement swapping on live commercial fiber as a carrier-compatible networking primitive [1, 2]. The four-branch structure documented here, and the classical channel that resolves it, are not abstract formalism — they are part of the operating mechanism of that demonstration. An implementation that obscures either would be technically untrustworthy. This one does not.

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References

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